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Continuous Integration with CodeBuild

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Introducing Today's Project!

In this project, I will demonstrate how to use AWS CodeVuild to automate the build process in the CI/CD Pipeline. Building means compressing our Web App's files into a single compressed file (Like a Zipfile) that we can later deploy. This project is going to focus on CodeBuild and Creating that compressed file automatically. Deploying it comes in the next project.

Key tools and concepts

Services I used were CodeBuild, CodeConnection, CodeArtifact, EC2, S3, GitHub and VS Code. Key concepts I learnt include the build process, Resolving git issues in the terminal, buildspec.yml file and using CodeConnection to connect AWS with GitHub.

Project reflection

This project took me approximately 4 hours including the documentation process. The most challenging part was resolving Git issues in the terminal and It was most rewarding to see the build succeed at the end.

This project is part four of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project in the comming few days.

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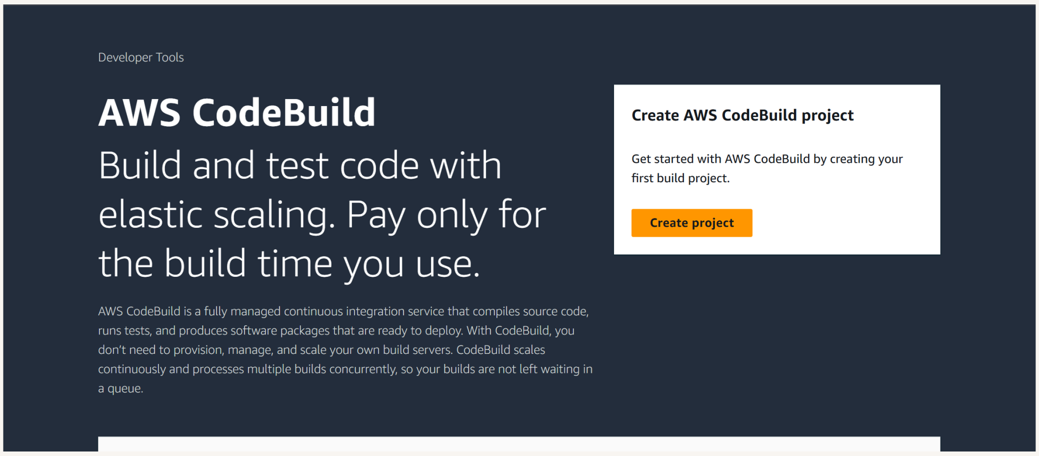
Setting up a CodeBuild Project

CodeBuild is a continuous integration service, which means it helps you to compile and package code (i.e turn your Web App Code into something computers can run then compress it all into a neat little package). Engineering teams use it because CI services automate this process for us - if CI does not exist, engineers would have to manualy compile and package code themselves before they deploy it.

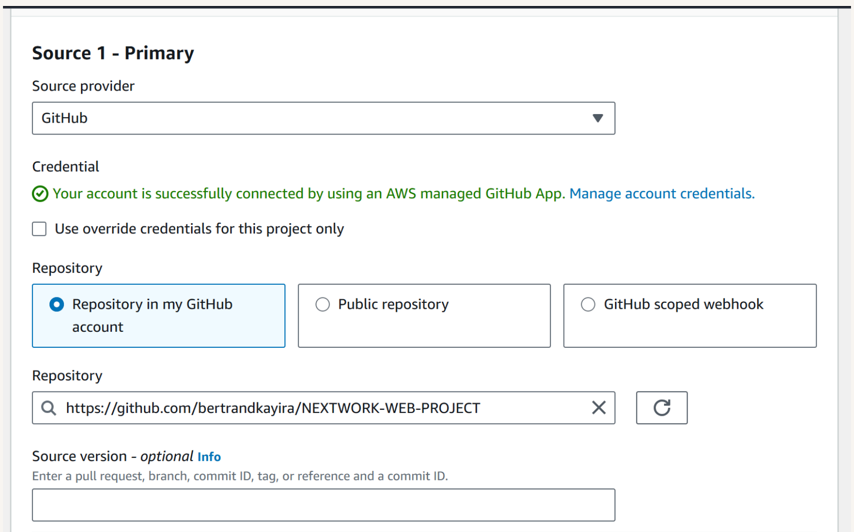
My CodeBuild project's source configuration means where the code lives and I selected GitHub! That's our Web App's Code Repository.



Connecting CodeBuild with GitHub

here are multiple credential types for GitHub, like GitHub App, Personal Access Tokens and OAuth app. I used GitHub App because it is the most simple and secure way to set up a connection between GitHub and AWS. AWS handles all the credentials (e.g the tokens) in the background for us. We simply need to login once to GitHub using a special AWS managed GitHub App.

The service that helped connect our AWS Environment with GitHub is AWS CodeConnection. CodeConnection also let's us connect our environment with other third party environments. In this case, it makes sure that we have secure connection to GitHub, so that our CodeBuild project can use our GitHub repository as the source with permission.



CodeBuild Configurations

Environment

My CodeBuild project's Environment configuration means the environment set up for the compute power (i.e the EC2 Instance) that will be compiling and compressing the files when we run our CodeBuild project. It includes settings like provisioning model, Compute, Operating System, image and Service Role. These settings determine how a new EC2 Instance will be spun up and conduct the build process. It's important here that the EC2 Instance has the correct permissions to everything it needs to work properly.

Artifacts

Build artifacts are files/resources that get created as a part od the CodeBuild process. They're important because they represent artifacts that needs to be used later on in CI/CD Pipeline - for example, our build process will create a compressed file that represents my Web App. I will need to use that compressed file to deploy the Web App Later on. To store them, I created an S3 bucket.

Packaging

When setting up CodeBuild, I also chose to package artifacts in a Zip file - this helps with package management as files are organized neatly in a single executable file. It also helps to compress the size of my compiled Web App.

Monitoring

For monitoring, I enabled CloudWatch Logs, which is a service that will note down commands that are run and any errors that happen. This is an extremely helpful for troubleshooting.

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buildspec.yml

My first build failed because CodeBuild could not find the buildspec.yml file in my source code root directory. A buildspec.yml file is needed because it tells CodeBuild how to run the build process.

The first two phases in my buildspec.yml file installs java give me access to CodeArtifact. The third phase in my buildspec.yml file compiles the Web App. The fourth phase in my buildspec.yml file packages the Web App into a single compressed artifact.

buildspec.yml

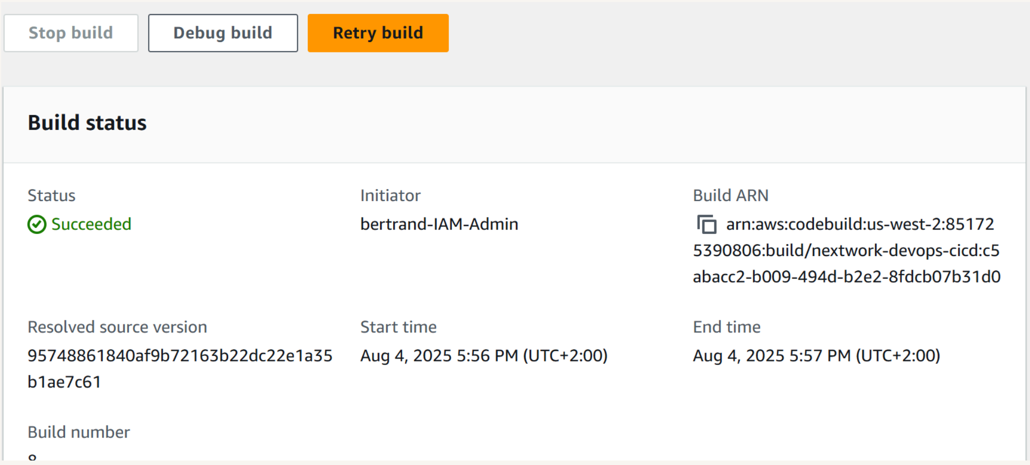
```yaml
version: 0.2

phases:
  install:
    runtime-versions:
      java: corretto8
  pre_build:
    commands:
      - echo Initializing environment
      - export CODEARTIFACT_AUTH_TOKEN=`aws codeartifact get-authorization-token --domain nextwork --domain-owner 123456789012 --regio

  build:
    commands:
      - echo Build started on `date`
      - mvn -s settings.xml compile
  post_build:
    commands:
      - echo Build completed on `date`
      - mvn -s settings.xml package
artifacts:
  files:
    - target/nextwork-web-project.war
  discard-paths: no
```

Success!

My second build also failed, but with a different error that said said I failed to run an error while executing the command to compile my Web App "mvn -s settings.xml compile." To fix this, I need to add a new policy to CodeBuild that grants CodeArifact permissions.

To resolve the second error, I attached a policy for accessing CodeArtifacts to my CodeBuild role. When I built my project again, I saw a success with the privildges to CodeArtifacts granted by role, CodeBuild now has access to the repository.

To verify the build, I checked Amazon S3. Seeing the artifact tells me that CodeBuild was a success. The build process compiled the Code from the source repository, compressed it into a file and stored that file into S3 bucket.



Stop build
Debug build
Retry build
Build status
Status
Succeeded
Initiator
bertrand-IAM-Admin
Build ARN
arn:aws:codebuild:us-west-2:851725390806:build/nextwork-devops-cicd:c5abacc2-b009-494d-b2e2-8fdcb07b31d0
Resolved source version
95748861840af9b72163b22dc22e1a35b1ae7c61
Start time
Aug 4, 2025 5:56 PM (UTC+2:00)
End time
Aug 4, 2025 5:57 PM (UTC+2:00)
Build number

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